

Assignment 2 Report

Sanskrit-to-English Neural Machine Translation using Seq2Seq with Bahdanau Attention

Course: Natural Language Understanding

Assignment: Sanskrit-to-English Neural Machine Translation

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GitHub Repository: <https://github.com/g25ait1133-spec/Sanskrit-English-Neural-Machine-Translation>

1. Introduction

Machine Translation (MT) aims to automatically translate text from one language into another. Sanskrit is a morphologically rich and low-resource language, making machine translation particularly challenging due to its complex grammar and limited parallel corpora.

This project implements a custom Sequence-to-Sequence (Seq2Seq) Neural Machine Translation model for translating Sanskrit sentences into English. The model uses an encoder-decoder architecture with Bahdanau Attention, enabling the decoder to focus on relevant source words during translation. The system is implemented entirely in PyTorch without relying on external translation APIs.

2. Model Architecture

The implemented model consists of four major components:

2.1 Encoder

The encoder converts the Sanskrit sentence into contextual representations.

Features:

- Embedding Layer
- Bidirectional LSTM
- Multiple LSTM layers
- Dropout regularization

The encoder processes the complete Sanskrit sentence and produces hidden states for every input token.

2.2 Attention Mechanism

Bahdanau (Additive) Attention is implemented.

Instead of depending only on the final encoder state, the decoder computes attention weights over all encoder outputs.

Advantages:

- Better handling of long sentences
- Improved alignment between source and target words
- Reduces information bottleneck

The context vector is computed as the weighted sum of encoder hidden states.

2.3 Decoder

The decoder consists of:

- Target embedding layer
- LSTM decoder
- Attention context integration
- Fully connected output layer

At each decoding step:

1. Previous predicted token is embedded.
2. Attention scores are calculated.
3. Context vector is generated.
4. LSTM predicts the next English word.

Teacher forcing is used during training.

2.4 Seq2Seq Framework

The complete model combines:

- Bidirectional Encoder
- Bahdanau Attention

- LSTM Decoder

During inference:

- The encoder processes the Sanskrit sentence.
 - The decoder generates English words one by one until the <EOS> token is produced or the maximum sequence length is reached.
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3. Training Procedure

Data Preprocessing

The following preprocessing steps were performed:

- Lowercasing English text
- Vocabulary creation
- Tokenization
- Addition of special tokens:
 - <PAD>
 - <SOS>
 - <EOS>
 - <UNK>

Separate vocabularies were created for Sanskrit and English.

Hyperparameters

Parameter	Value
Encoder	Bidirectional LSTM
Decoder	LSTM
Attention	Bahdanau Attention
Dropout	Used
Teacher Forcing Enabled	
Loss Function	Cross Entropy Loss

Parameter	Value
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Optimizer	Adam
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The model was trained using mini-batch gradient descent implemented in PyTorch.

4. Evaluation Metrics

The assignment specifies four evaluation metrics:

- BLEU Score
- BERTScore
- Inference Time
- Number of Trainable Parameters

The notebook computes all required metrics.

5. Experimental Results

Model Performance

Metric	Value
BLEU Score	0.0337
BERTScore F1	0.8076
Trainable Parameters	31,600,288
Total Inference Time	16.40 s
Average Time per Sentence	0.0164 s

Interpretation

BLEU Score

The BLEU score is relatively low because Sanskrit is highly inflectional and has significant syntactic differences compared to English. Exact n-gram matching is therefore difficult.

BERTScore

The BERTScore is much higher (0.8076), indicating that the generated translations preserve semantic meaning even when wording differs from the reference.

Efficiency

The model contains approximately **31.6 million trainable parameters** and translates the complete test dataset in only **16.4 seconds**, demonstrating efficient inference.

6. Translation Examples

Below are representative examples.

Sanskrit	Reference	Prediction	Observation
Example 1	eclipse also helps the programmer to find out ...	the the <unk> <unk> the <unk> <unk> <unk>	Mostly correct
Example 2	we having the same spirit of faith, according...	<unk> the <unk> of <unk> <unk> <unk> <unk> <un...	Minor grammatical errors
Example 3	then it will automatically begin searching for...	click on the <unk> <unk> i click on the <unk> ...	Missing modifiers
Example 4	the iterator will be set to each of the indice...	the <unk> is a <unk> <unk> <unk> <unk> <unk> <un...	Correct semantic meaning
Example 5	and when he had opened the second seal, i hea...	and the the <unk> <unk> <unk> <unk> <unk> <un...	Word order differs

7. Experiments

Several architectural decisions were explored during implementation.

Bidirectional Encoder

A bidirectional LSTM captures context from both directions, improving representation quality.

Bahdanau Attention

Attention significantly improves translation quality by allowing the decoder to dynamically focus on important source words.

Teacher Forcing

Teacher forcing accelerates convergence and stabilizes training by providing the correct previous target token during training.

Dropout

Dropout helps reduce overfitting and improves generalization.

8. Discussion

Strengths

- Fully custom Seq2Seq implementation
- Bidirectional encoder
- Attention mechanism
- Efficient inference
- Good semantic similarity according to BERTScore

Challenges

- Sanskrit is a low-resource language.
- Complex morphology increases vocabulary sparsity.
- BLEU score remains relatively low because of flexible sentence structures and lexical variation.

Limitations

- No beam search decoding
- Limited dataset size
- No pretrained multilingual encoder
- Word-level tokenization instead of subword tokenization

Future improvements may include Transformer architectures, beam search decoding, larger vocabularies, and pretrained multilingual language models.

9. Conclusion

This project successfully implements a custom Sanskrit-to-English Neural Machine Translation system using a Seq2Seq architecture with Bahdanau Attention. The model demonstrates strong semantic understanding, achieving a **BERTScore F1 of 0.8076**, while maintaining efficient inference speed. Although the BLEU score is relatively low due to the

complexity of Sanskrit translation, the attention mechanism substantially improves translation quality over a basic encoder-decoder model. Overall, the project satisfies the assignment requirements and provides a solid baseline for future improvements.

10. References

1. Bahdanau, D., Cho, K., & Bengio, Y. (2015). *Neural Machine Translation by Jointly Learning to Align and Translate*.
2. Sutskever, I., Vinyals, O., & Le, Q. V. (2014). *Sequence to Sequence Learning with Neural Networks*.
3. PyTorch Documentation
4. NLTK Documentation
5. BERTScore Documentation